Computer Networks Study Notes

Simple explanations • illustrated concepts • solved numericals • 30-mark practice paper

Your target: confident, complete answers

Based on all nine supplied PDFs (326 slides), covering lectures 1-10. These notes explain the taught topics in plain English. Practice questions and memory aids are added study support, not predictions of your actual paper. A score cannot be guaranteed, but you can use the checklist to find and fix weak areas.

How to use this guide

- **Understand:** read one topic and explain its diagram aloud without looking.
- **Recall:** close the guide and write definitions, comparisons and layer functions.
- **Apply:** solve the examples again with the answers hidden.
- **Test:** attempt the 30-mark paper, check the marking guide, then revise each lost mark.

Pages	What to study
2-3	Data communication, components and data representation
4	Network basics, transmission modes and line configuration
5-6	Topology diagrams, advantages, failures and mesh calculations
7	PAN, LAN, MAN, WAN and internetworks
8-10	OSI layers, encapsulation, TCP/IP, protocols and ports
11-12	Guided media: twisted pair, coaxial cable and optical fiber
13-14	Wireless media, propagation and frequency bands
15-17	FDM, WDM, TDM and worked calculations
18-19	Formula sheet, fast revision and short-answer practice
20-21	30-mark practice paper and marking guide
22	Source map and slide corrections

Scope: the later topics listed only in the course outline (such as detailed routing algorithms and CRC) are not taught in these nine files, so this guide does not invent full chapters for them.

01 Data communication

Communication means sharing information. **Telecommunication** means communication over a distance. **Data communication** means exchanging data between devices through a transmission medium.

A **data communication system** combines hardware (devices and cables) and software (programs and rules) to make that exchange possible. Local communication covers a limited area; remote communication connects distant locations.



Protocol = shared rules used by the communicating devices

Two views of the same system: the signal-processing chain above, and the five basic communication components below.

Component	Simple meaning	Example: sending a message
Message	The information being sent	Text, image, audio or video
Sender	Device that creates/sends data	Your phone
Receiver	Device that receives data	Your friend's phone
Medium	Path used by the signal	Radio waves, cable or fiber
Protocol	Agreed communication rules	Rules for format and exchange

Four signs of effective communication

Quality	Meaning	Easy example
Delivery	Reaches the correct destination	Your message reaches the intended friend.
Accuracy	Arrives without unwanted changes	A sent 1 does not become a 0.
Timeliness	Arrives when it is needed	A live call has little delay.
Jitter	Variation in packet arrival delay	Uneven arrivals make audio/video choppy.

Remember D-A-T-J

Delivery = right place. Accuracy = right data. Timeliness = right time. Low jitter = steady timing. Delay and jitter are different: a constant delay can have low jitter.

01 Data representation

Data type	How it is represented	Remember
Text	Characters are mapped to bit patterns using an encoding.	ASCII and Unicode are examples.
Numbers	Numeric values can be stored in binary.	Decimal 5 = binary 101. Text "5" is encoded differently.
Images	A grid of pixels; each pixel has a colour or brightness value.	More pixels or more bits per pixel usually need more storage.
Audio	Sound is naturally continuous; digital audio stores samples as numbers.	A microphone first produces an electrical signal.
Video	A sequence of image frames represents motion.	Digital video often includes an audio stream.

Pixels, colours and storage

• **1 bit per pixel:** 2 possible values, such as black and white.

• **2 bits per pixel:** 4 levels, e.g. 00, 01, 10, 11. In general, b bits represent 2^b values.

• **RGB:** red, green and blue components combine to make colours. The slides also mention cyan, magenta and yellow (CMY).

Small worked example

A 100 × 100 image at 1 bit/pixel needs 10,000 bits = 1,250 bytes before headers/compression. At 2 bits/pixel it needs 20,000 bits = 2,500 bytes. Formula: width × height × bits per pixel.

The detailed digital communication chain

Stage	Purpose
Format	Represent the original message in a form suitable for processing.
Source encode	Compress: remove unnecessary redundancy.
Encrypt	Hide the message from unauthorized readers.
Channel encode	Add controlled redundancy to detect/correct transmission errors.
Multiplex	Combine several streams on one link.
Modulate / signal	Represent data as a waveform suitable for the channel.
Spread / multiple access	Spread spectrum for interference resistance; share resources among users.
Transmit / receive	Send the signal; the receiver reverses the applicable processing.

The lecture marks several stages as optional; the chain is a model, not a requirement that every network must perform every step. Unicode is not always 32 bits: UTF-8, UTF-16 and UTF-32 are different encodings. See page 22.

02 Network fundamentals

A **network** is a group of connected devices that exchange information or share resources. A **node** is a participating device; a **link** joins adjacent nodes; a **path** is the complete route across one or more links.

Term	Short explanation
Session	An ongoing communication dialogue between applications/users.
Circuit	A communication path or channel used to carry data.
Packetizing	Splitting a message into smaller packets; packet sizes need not always be fixed.
Routing	Choosing a path from source to destination.
Distributed processing	Dividing a task among multiple computers.

Three transmission modes

Mode	Direction	Example
Simplex	A → B only	Radio broadcast; keyboard input as a simplified example.
Half-duplex	A → B OR B → A; take turns	Walkie-talkie.
Full-duplex	A → B AND B → A together	Telephone conversation.

Why build networks?

Share files, printers, storage and software services; communicate by email; centralize backups; control access; distribute work. A separate direct cable between every pair becomes costly as the number of devices grows.

Network criterion	What to explain in an answer
Performance	Transit time = travel time between devices. Response time = time from request to reply. Depends on users, medium, hardware and software.
Reliability	How often failures occur, how quickly recovery happens, and how well the network survives a major disruption.
Security	Prevent unauthorized access and damage; use policies and recovery procedures after breaches or data loss.

Line configuration: who shares the link?

Point-to-point: one dedicated link between two devices; its capacity is for those endpoints. **Multipoint / multidrop:** more than two devices share a link. Sharing may be simultaneous (spatial) or by turns (temporal). A link may be wired or wireless.

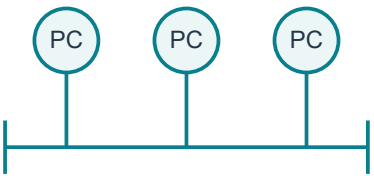
Do not mix these three classifications

Mode = direction of communication. Line configuration = devices sharing a link. Topology = arrangement of the whole network.

03 See the topologies

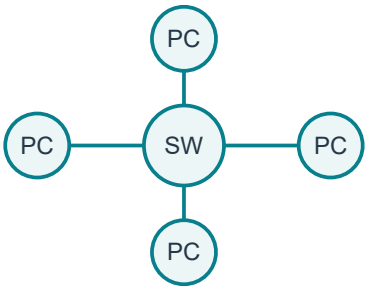
Topology describes how nodes and links are arranged. Physical topology is the actual layout; logical topology describes how data flows. SW below means a central switch/controller.

BUS

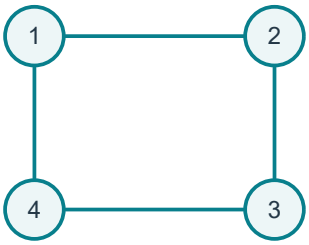


Shared backbone

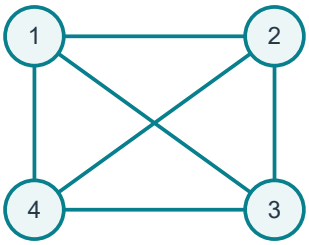
STAR



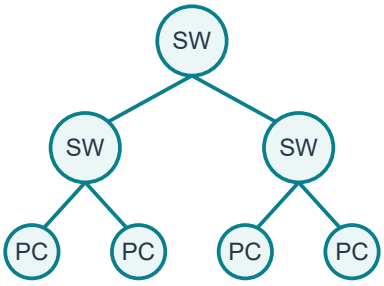
RING



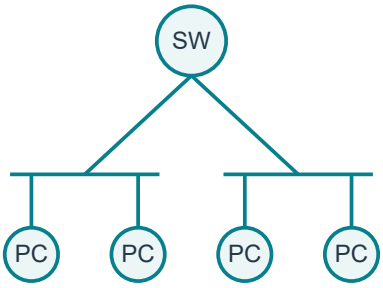
FULL MESH



TREE



HYBRID



Topology	How it works
Bus	All nodes attach to a shared backbone using drop lines/taps. A signal spreads along the cable; the addressed receiver accepts it.
Star	Every node has a separate link to a central hub or switch. Traffic between nodes passes through that controller.
Ring	Each node connects to two neighbours. In the lecture's token ring, a circulating token gives permission to transmit.
Full mesh	Every node has a dedicated link to every other node. A partial/hybrid mesh has some extra links but is not fully connected.
Tree	A hierarchy of branches, often formed from connected stars.
Hybrid	A combination of topologies, such as the star backbone with bus segments shown above.

Diagrams are conceptual. A ring need not always send clockwise. Token control prevents simultaneous competing transmissions in the token-ring model; a circular shape alone does not guarantee no collisions.

03 Compare and calculate

Topology	Advantages	Limitations / failures
Bus	Low cable cost; simple for small installations.	Backbone faults can disrupt the whole segment; shared traffic and collisions; difficult fault finding.
Star	Easy to add nodes and isolate faults; one endpoint link failure usually affects only that node.	Central controller failure affects its star; more cable than a bus.
Ring	Orderly token access; no central controller required in the basic model.	A break or failed forwarding node can disrupt a simple ring; changes may interrupt service.
Full mesh	Redundant paths; dedicated links; easier link-fault isolation.	Many cables and ports; expensive and difficult to install.
Tree	Easy to organize branches and expand hierarchically.	An upstream device/link failure can disconnect an entire branch.
Hybrid	Combines useful properties of different layouts.	Design and troubleshooting can be more complex.

Bus components and behaviour

NIC: network interface card. **T-connector/tap:** attaches a node to the bus. **Drop line:** joins node to backbone. **Barrel connector:** joins cable sections. **Terminators:** absorb signals at both ends to prevent reflections.

The lecture describes shared Ethernet using **CSMA/CD**: listen before sending, detect collisions, then retry. It applies to classic shared/half-duplex Ethernet, not every network with a bus-like drawing. A hub repeats traffic; a switch forwards based on learned destination information.

Full-mesh numericals

Memorize these formulas

For n devices: **links = n(n - 1)/2**; **ports per device = n - 1**; **total port endpoints = n(n - 1)**. Assumes one bidirectional link per pair. Divide by 2 for links because counting from every device counts each link twice.

Example 1: 6 devices → 6 × 5 / 2 = **15 links**; 5 ports/device; 30 total port endpoints.
Example 2: 7 devices → 7 × 6 / 2 = **21 links**. Adding the seventh device to a six-node full mesh requires **6 new links**.

Failure trap: an ordinary endpoint failure does not necessarily destroy a bus. A backbone break or a device fault that disrupts the shared medium can. Avoid the slide's blanket claim that every computer failure stops the bus.

04 Categories of networks

Type	Typical scope	Simple example
PAN: Personal Area Network	Devices around one person	Phone connected to personal accessories.
LAN: Local Area Network	A room, building or limited campus area	University computer lab.
MAN: Metropolitan Area Network	A city/metropolitan region	Several office LANs across one city.
WAN: Wide Area Network	Large region, country or continents	A bank connecting branches nationwide.

Scale memory aid

Person → building/campus → city → country/world.
PAN → LAN → MAN → WAN. These are useful scope categories, not universal exact distance limits. A home network shared by a household is often called HAN and is commonly a LAN.

LAN access: broadcast versus switched

Broadcast/shared LAN: a transmission can reach all stations on the shared medium; an access method coordinates senders. **Switched LAN:** a switch forwards frames toward the destination port/segment. Broadcast traffic can still exist in a switched LAN.

Names in the slides: Ethernet, Fast Ethernet, Gigabit Ethernet, 10G Ethernet, wireless LAN, Token Ring, FDDI, ATM LAN and other historical technologies. Treat the slide speed figures as examples, not fixed speed limits for LANs.

WAN classifications

Classification	Meaning
Enterprise WAN	A WAN used by one organization, e.g. the lecture's HBL/NADRA examples. Its links may be leased from a provider.
Global WAN	A network with very broad/global reach; the slides cite Internet and PSTN.
Point-to-point WAN	A dedicated connection between two endpoints, often connecting a site to a provider.
Switched WAN	A provider network with intermediate switching nodes connecting many sites. Slide examples: X.25, Frame Relay and ATM.

Networks grouped by purpose

SAN: Storage Area Network, dedicated to storage access. **EPN:** Enterprise Private Network, connects an organization's resources. **VPN:** Virtual Private Network, provides private logical connectivity over another network, often using encrypted tunnels.

Internetwork versus Internet

An **internetwork** joins two or more networks, normally through routers. The **Internet** is the worldwide interconnected network of networks using the Internet protocol suite. Example: lab LAN → router → WAN → router → another campus LAN.

05 OSI: the seven layers

OSI = Open Systems Interconnection. It is a conceptual reference model from ISO. Layers divide a complex job into manageable functions. Each layer uses the layer below and offers services to the layer above through a defined interface.

Layer	Main job	Key words to write
7 Application	Provides network services to applications/users.	Email, file transfer, directory services, virtual terminal.
6 Presentation	Makes data representation usable to the receiver.	Translation, encryption/decryption, compression.
5 Session	Manages the communication dialogue.	Establish, maintain, end; dialogue control; checkpoints.
4 Transport	Process-to-process delivery.	Ports; segmentation/reassembly; connection, flow and error control.
3 Network	Source-host to destination-host packet delivery.	Logical/IP addressing; routing and forwarding.
2 Data link	Node-to-node / hop-to-hop frame delivery.	Framing; MAC addresses; access, flow and error control.
1 Physical	Transmits bits as physical signals.	Encoding; interfaces; bit timing; data rate; medium.

Learn the order both ways

Top to bottom: **All People Seem To Need Data Processing**.
Application → Presentation → Session → Transport → Network → Data link → Physical.
Bottom to top: **Please Do Not Throw Sausage Pizza Away**.

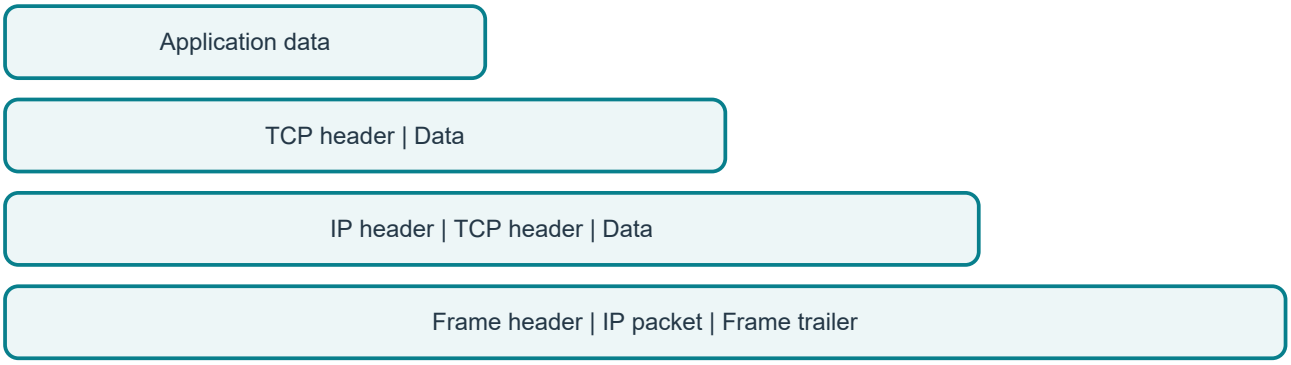
Peer processes and interfaces

Peers: matching layers on different machines logically communicate using a protocol. **Interface:** the service boundary between adjacent layers on one machine. Data physically travels down the sender's stack, through the medium, and up the receiver's stack.

Lecture grouping: layers **1-3 support the network**; layers **5-7 support the user**; layer **4 connects the groups**. Modularity makes design, interoperability and troubleshooting easier.

Do not write that OSI was created in 1947. That date in the slide refers to ISO's founding, not the seven-layer model. Also, reliability depends on the actual protocol: not every transport protocol guarantees delivery.

05 Follow a message



Sender adds control information; receiver removes it in reverse order.

Encapsulation means adding control information as data moves down the stack. **Decapsulation** removes it at the receiver. Typical names: data → TCP segment (or UDP datagram) → IP packet → frame → bits. A frame commonly has both a header and a trailer.

The three delivery scopes

Layer	Question answered	Address
Transport	Which application/process should receive it?	Port number
Network	Which host/network should it reach?	Logical/IP address
Data link	Which local/next-hop interface receives this frame?	MAC address on an Ethernet link

Example: your browser sends data to a remote web server. The transport port identifies the service. The IP destination identifies the server. The first Ethernet frame is addressed to the next-hop router if the server is outside the local network. Routers forward the packet and create suitable link-layer frames for the next link.

Functions that students often confuse

Function	Simple explanation
Segmentation / reassembly	Transport breaks an application message into manageable pieces and rebuilds the stream/message as appropriate.
Flow control	Prevents a sender from overwhelming a receiver.
Error control	Detects errors and may use retransmission/correction, depending on the protocol.
Routing / forwarding	Routing chooses paths; forwarding sends an arriving packet toward its next hop.
Framing / access control	Framing marks a data unit; access control decides who may use a shared medium.
Synchronization	Session checkpoints help resume work; physical bit synchronization keeps sender and receiver timing aligned.

Data link sublayers: LLC (Logical Link Control) supports link services; MAC (Media Access Control) handles medium access and MAC-related framing/addressing in IEEE LANs.

Physical-layer checklist: interface/medium characteristics, bit representation, data rate, bit synchronization, line configuration, topology and transmission mode. The slides also place baseband/broadband and multiplexing here.

06 TCP/IP made simple

TCP/IP is the Internet protocol suite, developed from the DARPA/DoD networking work and standardized through the IETF. It is both a practical protocol family and the basis of a layered model.

OSI layers	TCP/IP: four-layer view	Five-layer teaching view
Application + Presentation + Session	Application	Application
Transport	Transport	Transport
Network	Internet	Network / Internet
Data link + Physical	Link / Network interface / Host-to-network	Data link + Physical (separate)

OSI has 7 reference layers; TCP/IP is commonly shown with 4 layers, or with 5 when the link layer is split. Session and presentation are not separate TCP/IP layers; their needed functions are handled by applications and supporting protocols. TCP connection management alone is not the whole OSI session layer.

Feature	TCP	UDP
Service	Connection-oriented reliable byte stream.	Connectionless datagrams.
Reliability	Uses acknowledgements/retransmission and ordering.	No built-in delivery or ordering guarantee.
Control	Provides flow and congestion control.	No equivalent built-in reliability/flow-control service.
Useful when	Correct, ordered delivery matters.	Timeliness or low protocol overhead matters; applications may add controls.

Ports and sockets

Ports distinguish application endpoints and let many applications share the network. A **socket endpoint** is commonly identified by an IP address and port; a connection also involves the peer endpoint and transport protocol.

Service	Port(s) in your slides	Purpose
FTP	21 control; 20 classic active-mode data	File transfer
Telnet	23	Remote terminal
SMTP	25	Mail transfer
HTTP	80	Web communication
POP3	110	Mail retrieval

IP: addressing and packet delivery across networks. **ARP:** IPv4 address to local link/MAC address. **RARP:** historical reverse mapping. **ICMP:** IP control/error reporting. **IGMP:** IPv4 multicast group membership. The lecture groups these with Internet-layer support; ARP operates at the link/Internet boundary.

07 Guided media: copper

A **transmission medium** carries signals between devices. **Guided media** confine/direct signals along a cable. **Unguided media** carry electromagnetic waves through free space. Media are used by the physical layer; “layer 0” is an informal expression, not an extra official OSI layer.

Twisted pair: two insulated copper conductors

The receiver detects a voltage difference between the wires. Twisting exposes both wires more similarly to external noise, helping cancel shared interference. **Crosstalk** is unwanted coupling from nearby signal wires. Twisting reduces noise; it does not remove every possible disturbance.

Type	Construction	Trade-off
UTP	Unshielded twisted pair.	Cheap, flexible and easy to install; more exposed to interference.
STP	Twisted pair plus foil/braided shielding.	Better interference protection when installed correctly; higher cost and installation demands.

Connector: the slides call the common Ethernet plug RJ-45. **Uses:** telephone wiring and wired LAN links. Shielding and twists help protect the signal in different ways.

Cable category table: recognize the lecture labels

Category	Lecture table: rate (Mbps)	Lecture association
1 / 2	< 0.1 / 2	Telephone / T-1 lines
3 / 4	10 / 20	LAN / Token Ring
5 / 5E	100 / 125	LAN; 5E improves interference/crosstalk performance
6 / 7	200 / 600	Higher-performance cabling; Cat 7 has shielding

These are the slide's historical figures, not dependable modern maximum data rates. In particular, Cat 5e supports Gigabit Ethernet; do not memorize 125 Mbps as its general speed limit. See page 22.

Coaxial cable: layers from inside to outside

Conductor → insulation → metallic shield → jacket

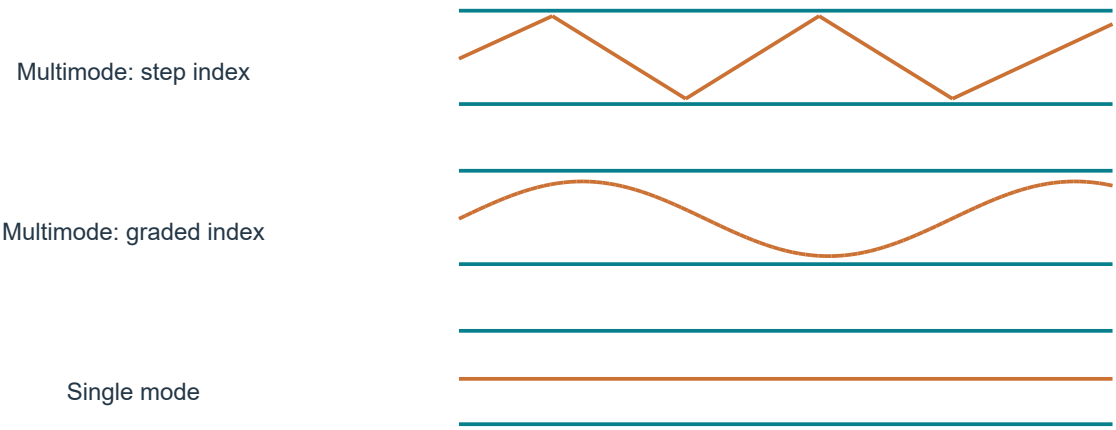
The centre conductor carries the signal. The dielectric insulation separates it from the outer conductor. The metallic outer conductor provides shielding and a return path. The jacket protects the cable.

Uses: cable television, historical telephone trunks, classic coaxial Ethernet. **BNC connector:** cable-to-device connection; **BNC T:** branch to a device; **terminator:** prevents reflection. RG labels specify cable construction. The slides associate RG-59 with cable TV and RG-58 with thin Ethernet. Their RG-11/50-ohm entry is incorrect; see page 22.

07 Guided media: optical fiber

Optical fiber carries light through a glass/plastic **core** surrounded by **cladding**. The core has a higher refractive index than the cladding. With a suitable incidence angle, **total internal reflection** keeps light inside the core.

Refraction: light changes direction when entering a material with a different refractive index. **Critical angle:** the incidence angle for which the refracted ray is at 90° to the normal. Above this angle, light is totally reflected when travelling from higher to lower refractive index.



Simplified ray sketches: step-index has an abrupt index change; graded-index changes gradually; single-mode carries one guided mode. “Mode” describes a propagation pattern, not a separate user.

Fiber type	Key idea	Result
Multimode step-index	Several paths through a uniform-index core.	Different path delays cause modal dispersion (pulse spreading).
Multimode graded-index	Index gradually decreases from centre to edge.	Different propagation speeds reduce delay differences.
Single-mode	Small core supports one guided mode.	Avoids intermodal dispersion; useful for long-distance links.

Sizes in the lecture: 50/125, 62.5/125 and 100/125 micrometres for multimode; 7/125 for single-mode. The first number is core diameter, the second cladding diameter. These are examples, not the only possible fiber sizes.

Construction: core + cladding + protective coating + strength members (e.g. Kevlar) + outer jacket. **LED:** less focused, often lower-cost short-distance source. **Laser/ILD:** focused source suited to longer-distance/high-performance links. **Connectors:** SC (push/pull), ST (bayonet), MT-RJ.

Advantages: high capacity, low attenuation, immunity to electromagnetic interference, low weight, resistance to corrosion and harder tapping. **Limitations:** installation/splicing skill, equipment cost and bend/handling sensitivity. Duplex operation needs appropriate paths or wavelengths; fiber is not inherently restricted to one-way communication.

Applications in the slides: backbone networks, SONET, fiber LANs, and hybrid fiber/coax cable-TV systems. Attenuation = loss of signal strength; dispersion = pulse spreading. They are different problems.

08 Wireless transmission

Unguided communication sends electromagnetic energy through free space rather than a cable. An antenna transmits or receives the signal. Wireless signals may be overheard within range, so security depends on more than simply reaching the intended receiver.

Propagation	Simple picture	Key fact
Ground	Signal follows the Earth's surface.	Low-frequency waves can travel around Earth's curvature.
Sky	Signal goes upward and returns via the ionosphere.	Can reach beyond the horizon; depends on frequency and atmospheric conditions.
Line of sight	Direct path between antennas.	Obstacles and Earth's curvature can block the path.

Compare the three lecture categories

Type	Lecture frequency convention	Characteristics / uses
Radio	3 kHz to 1 GHz	Often omnidirectional; antennas need not face each other. Broadcast radio/TV, maritime radio, paging. Some bands penetrate walls and travel far.
Microwave	1 GHz to 300 GHz	Directional links commonly need alignment and line of sight. Terrestrial links, satellite, cellular and WLAN applications.
Infrared	About 300 GHz to 400 THz	Short-range optical wireless; opaque walls block it. Remote controls and IrDA; sunlight may interfere.

The infrared slide says 400 GHz on slide 23 but refers to 400 THz on slide 24. Use THz for the approximate upper boundary. Radio/microwave naming boundaries vary; above are the course conventions.

Microwave antennas and satellite links

Parabolic dish: concentrates incoming parallel waves at a focus; reverses the process when transmitting. **Horn antenna:** a flared opening that directs/collects radio energy; can be used as a dish feed. **Terrestrial microwave:** ground stations communicate directly; repeaters extend a route. **Satellite:** receives an uplink and retransmits a downlink, acting as a high-altitude relay.

Avoid absolute claims

The lecture uses omnidirectional radio versus directional microwave as a helpful comparison. Actual radiation pattern depends on the antenna. Likewise, wall penetration and weather effects depend on frequency and material, not just the category name.

One-to-one = unicast. One-to-selected-many = multicast. One-to-all in the relevant network = broadcast. The slides use radio broadcasting as a one-to-many illustration; not all one-to-many transmission is IP multicast.

08 Bands and media selection

Band	Range	Lecture examples
VLF: very low frequency	3-30 kHz	Long-range radio navigation
LF: low frequency	30-300 kHz	Radio beacons/navigation
MF: medium frequency	300 kHz-3 MHz	AM radio
HF: high frequency	3-30 MHz	Citizens band; ship/aircraft radio
VHF: very high frequency	30-300 MHz	FM radio, VHF TV
UHF: ultra-high frequency	300 MHz-3 GHz	Cellular, UHF TV, satellite
SHF: super-high frequency	3-30 GHz	Satellite communication
EHF: extremely high frequency	30-300 GHz	Radar, satellite

Pattern: every row spans a factor of 10. Ground propagation is associated with low frequencies; sky-wave operation is important in MF/HF; VHF and higher are generally taught as line-of-sight. Real propagation modes overlap, so the slide table is simplified.

Five factors for choosing a medium

Factor	Question to ask
Cost	What do the cable, equipment, installation and maintenance cost?
Speed / capacity	Can the medium and equipment carry the required data rate?
Attenuation	How much strength is lost across the distance? Are repeaters needed?
EM interference	How much external electromagnetic noise affects the signal?
Security	How easily can a signal be intercepted or a cable tapped?

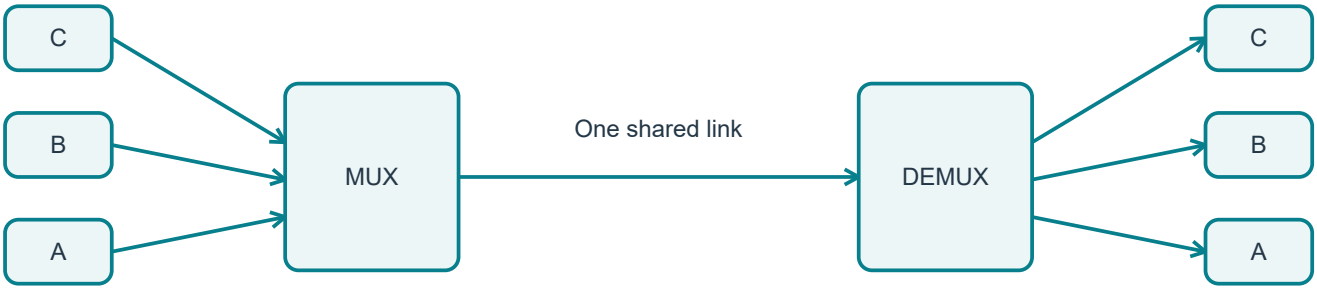
Choice	Good fit	Main trade-off
UTP	Affordable wired links inside a building.	More vulnerable to interference than properly shielded cable.
STP / coax	Electrical links needing shielding.	Extra bulk/cost; still subject to electrical limitations.
Fiber	High-capacity backbones or noisy environments.	Specialized installation and optical equipment.
Wireless	Mobility or places where cabling is difficult.	Interference, coverage, obstacles and interception risk.

Answer a scenario with a reason

“Use fiber between buildings because it supports high capacity, has low attenuation and resists electromagnetic interference.” A named medium plus a relevant reason scores better than a bare list of media.

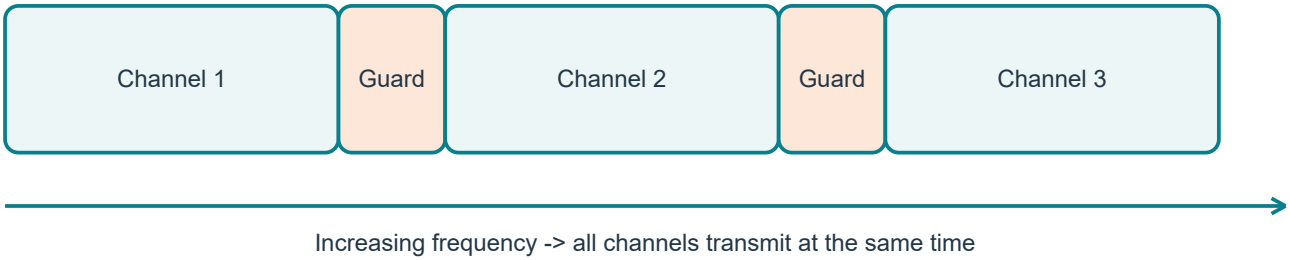
09-10 Multiplexing: FDM + WDM

Multiplexing allows several signals to share one link. A **MUX** combines inputs; a **DEMUX** separates them. The physical link is the shared resource; each allocated part is a channel. Sharing reduces the need for a separate physical link per source.



FDM: divide the frequency range

Frequency Division Multiplexing gives each signal a different carrier-frequency band. All channels can transmit at the same time. The receiver filters the required band and recovers its signal. **Guard bands** are unused frequency gaps that prevent adjacent-channel overlap/interference.



FDM is an analog multiplexing technique; digital data can still use it after suitable modulation. Applications in the slides include AM/FM broadcasting, TV channels and first-generation cellular systems. Specific broadcast bandwidths depend on the system/region.

Worked example: three voice channels

Each needs 4 kHz. With no guard bands, total bandwidth = 3 × 4 = **12 kHz**. In the 20-32 kHz range: channel 1 = 20-24, channel 2 = 24-28, channel 3 = 28-32 kHz.

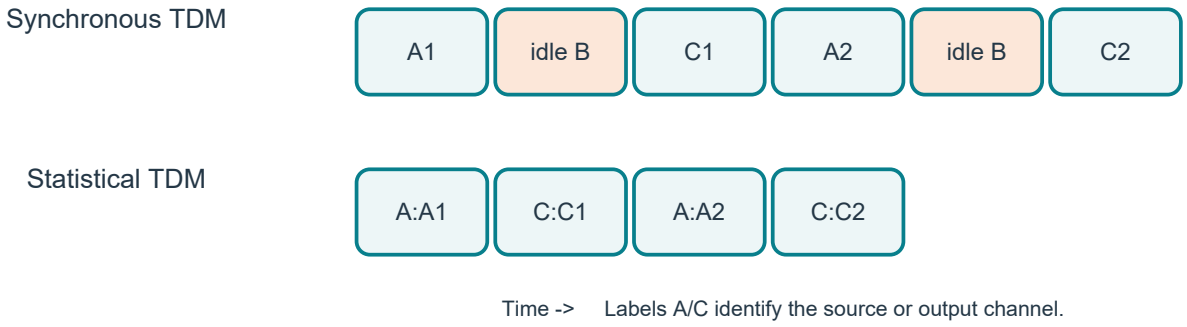
WDM: divide optical wavelengths

Wavelength Division Multiplexing combines light signals of different wavelengths on one optical fiber, then separates them at the receiving end. Think of different “colours” travelling through the same fiber. The lecture's prisms illustrate combining/separating wavelengths conceptually.

FDM versus WDM: both separate channels by frequency/wavelength; WDM uses optical signals in fiber. It uses fiber capacity more efficiently. WDM is not a type of time-slot sharing.

09-10 TDM: share the time

Time Division Multiplexing divides transmission time into slots. Sources take turns using the shared link. A slot may hold a bit, character or block. Analog sources can be digitized first. **Interleaving** means taking units from inputs in turn and placing them into the outgoing stream.



Example: source B is idle. Synchronous TDM still reserves B's slots; statistical TDM sends active channels and includes identification. The displayed units are illustrative, not literal fixed packet formats.

Feature	Synchronous TDM	Statistical / asynchronous TDM
Allocation	Fixed, recurring slots for each input.	Slots assigned to inputs with data.
Idle source	Its reserved slot is empty/wasted.	Skipped; capacity can serve active inputs.
Identification	Slot position identifies the channel.	Channel/address information is needed.
Capacity	Provisioned for scheduled input rates plus overhead.	Can exploit non-simultaneous activity; buffering handles bursts.
Trade-off	Simple and predictable; wastes idle slots.	More efficient for bursty use; identification overhead and possible queueing delay.

Frame versus slot

A **TDM frame** is one complete round of slots. If n equal-rate sources each contribute one unit, a frame has n slots. At output, each unit must be sent n times faster (ignoring overhead), so all n sources keep up.

The key timing relationships

Let each input send r bits/s and each input unit contain k bits.
Input-unit duration = k/r seconds.
Output rate = n × r bits/s (no overhead).
Output-slot duration = k/(n × r) seconds.
Frame duration = k/r seconds; frame size = n × k bits.

Lecture examples of synchronous multiplexing systems: T-1, ISDN and SONET. The “frame” in this section is a TDM scheduling structure; do not automatically treat it as an Ethernet frame.

09-10 Solved numericals

1. FDM with guard bands

Question: 5 channels each need 100 kHz; adjacent channels need 10 kHz guards. Minimum total bandwidth?
Formula: B = n × b + (n - 1) × g.
Substitute: B = 5 × 100 + 4 × 10 = **540 kHz**.
Why four guards? Five channels have four internal gaps; no external edge guards were specified.

2. Synchronous TDM timing

Question: 3 inputs, each 1 kbps, bit-by-bit interleaving, no overhead.
Input bit duration = 1/1000 s = **1 ms**.
Output rate = 3 × 1000 = **3000 bps = 3 kbps**.
Output-slot duration = 1/3000 s = **0.333 ms**.
Frame size = **3 bits**; frame duration = 3/3000 s = **1 ms**.
Frame rate = **1000 frames/s**.

3. What if input rates differ?

Technique	How it works	Example
Multilevel multiplexing	Combine lower-rate inputs first, then multiplex equal-rate groups.	20 + 20 → 40 kbps; combine with three other 40-kbps inputs → 160 kbps.
Multiple-slot allocation	Give faster sources more slots per frame.	A 50-kbps input gets twice the slots of a 25-kbps input.
Pulse stuffing / padding	Insert dummy bits into slower streams to match a chosen rate.	Raise 46 to 50 kbps with 4 kbps dummy bits, then multiplex.

Pulse stuffing here means rate matching; it is not the same purpose as inserting bits to prevent a framing delimiter from appearing in data.

4. Statistical TDM addresses

To identify N output channels, use at least **ceiling($\log_2 N$) bits**. Pick the smallest b for which $2^b \geq N$.
For 8 channels: $2^3 = 8 \rightarrow$ **3 bits**.
For 10 channels: 2^3 is too small, $2^4 = 16 \rightarrow$ **4 bits**.

Numerical answer format

Write: given values → correct formula → substitution → result with units. Before calculating, check whether the question specifies guard bands, framing bits, unequal rates or multi-bit slots.

Fast revision: one-page recall

Must remember	Recall line
Communication	Message, sender, receiver, medium, protocol. Quality: delivery, accuracy, timeliness, jitter.
Network	Performance, reliability, security. Distributed processing divides a task across computers.
Direction	Simplex: one way. Half-duplex: both ways by turns. Full-duplex: both ways together.
Topology	Bus: backbone. Star: centre. Ring: two neighbours. Mesh: many direct links. Tree: hierarchy. Hybrid: combination.
Scale	PAN → LAN → MAN → WAN. Internetwork = connected networks.
OSI	Application, Presentation, Session, Transport, Network, Data link, Physical.
Addresses	Transport: port. Network: IP. Data link: MAC.
TCP/IP	4-layer model; 5-layer teaching view. TCP reliable stream; UDP connectionless datagrams without built-in reliability.
Media	Copper: electrical signals. Fiber: light. Unguided: free-space electromagnetic waves.
Multiplexing	FDM: frequency bands. WDM: optical wavelengths. TDM: time slots.

Formula strip

Quantity	Formula / conversion
Full-mesh links	$n(n - 1)/2$
Ports per full-mesh node	$n - 1$
FDM total bandwidth	Sum of channel bandwidths + sum of guard bands
Equal-rate TDM, no overhead	$R = n \times r$; frame size = $n \times k$ bits; frame rate = r/k
TDM with f overhead bits/frame	$R = (n \times k + f) \times (r/k)$ bits/s
Statistical address bits	Smallest b with $2^b \geq N$
Image storage, uncompressed	width × height × bits per pixel
Units	1 byte = 8 bits; 1 kbps = 1000 bps; 1 ms = 0.001 s; 1 MHz = 1000 kHz

TDM formulas assume equal input rates r, k bits per source per frame and fixed framing overhead f. Frequency bandwidth (Hz) and data rate (bits/s) are different quantities; do not interchange their units.

Short-answer practice

Cover the answer column. Say each answer aloud, then write it in your own words. Expand the number of points to match the marks available.

Question	Answer you should be able to produce
Why are protocols needed?	They give devices common rules for interpreting and exchanging data; a physical connection alone is insufficient.
Delay versus jitter?	Delay is travel/waiting time; jitter is variation in delay.
Why twist the wires?	To expose the pair similarly to noise, allowing differential reception to reject much of it.
Why terminate a bus?	To absorb the signal at cable ends and prevent reflections.
Why does mesh cost more?	Links grow as n(n - 1)/2; every node needs n - 1 ports.
Why use layers?	To divide functions, define interfaces and support modularity/interoperability.
Presentation versus session?	Presentation handles representation; session controls dialogues and checkpoints.
Transport versus network?	Transport delivers to processes; network delivers packets between hosts across networks.
Network versus data link?	Network uses logical addressing/routing across networks; data link sends frames to the next hop.
What does a port identify?	A transport/application endpoint on a host, allowing multiple applications to share networking.
Why is fiber immune to EM interference?	It carries light in a dielectric medium rather than electrical current in a metal conductor.
Step-index versus graded-index?	Step-index changes refractive index abruptly; graded-index varies it gradually to reduce modal delay differences.
Why use guard bands?	To keep adjacent FDM channels from overlapping/interfering.
Why waste slots in synchronous TDM?	Slots are reserved even when a source has no data.
Why identify statistical TDM slots?	Their positions are not permanently assigned to specific channels.

A strong short-answer pattern

Definition → how it works → example or consequence.
For a comparison, use matched criteria. For a diagram, label nodes, links and direction. For a numerical, include units and assumptions.

Final self-check: Can you redraw all six topologies, write all OSI functions, compare the media and solve a fresh TDM problem without notes? Any “no” becomes your next revision task.

30-mark practice paper

Suggested practice time: 45 minutes. Attempt all questions with your notes closed. Suggested timing is for practice only; follow your actual exam instructions on the day.

Section A - short answers [10 marks]

Q1. [2] Name the five basic components of a data communication system, and define a protocol.

Q2. [2] Distinguish half-duplex from full-duplex and give one example of each.

Q3. [2] Distinguish a MAN from a WAN and give a suitable example of each.

Q4. [2] Explain why a bus needs terminators and why a star depends on its central controller.

Q5. [2] Explain why optical fiber uses a core and cladding, and name one advantage of fiber over copper.

Section B - concepts and diagrams [12 marks]

Q6. [4] Draw the seven OSI layers in order. State the main functions of the transport, network and data link layers. Identify the addressing associated with each of those three layers.

Q7. [4] Compare FDM, WDM and synchronous TDM by explaining what is divided. Include one labelled sketch of any one method. Explain why statistical TDM needs channel/address information.

Q8. [4] Compare UTP, STP and optical fiber in terms of construction and interference. Recommend a medium for a high-capacity inter-building link in an electrically noisy environment and justify it.

Section C - numericals [8 marks]

Q9. [2] A full-mesh network has 8 nodes. Find the number of links and the number of ports required at each node.

Q10. [2] Four FDM channels each require 50 kHz. There is a 5-kHz guard band between adjacent channels and no edge guards. Find the total bandwidth.

Q11. [4] Four input lines each send 2 kbps. Synchronous TDM takes 1 bit from each input per frame, without overhead. Find:
(a) output data rate, (b) input-bit duration, (c) output-slot duration, (d) frame duration.

Before looking at the answers

Check that your marks add to 30. Circle any guessed answer. In numericals, verify Hz versus bits/s, milliseconds versus seconds, and links versus ports. Review your diagrams for clear labels.

Practice-paper marking guide

Q	Model answer and suggested marks
1 [2]	Message, sender, receiver, medium, protocol (1 for the complete list). Protocol = agreed rules governing communication (1).
2 [2]	Half-duplex: both directions, one at a time; walkie-talkie (1). Full-duplex: both simultaneously; telephone call (1).
3 [2]	MAN: city-scale, e.g. office LANs across a city (1). WAN: larger geographical distances, e.g. nationwide bank network (1).
4 [2]	Terminators absorb signals to stop reflections (1). The star controller relays communication, so its failure stops communication through that star (1).
5 [2]	Higher-index core plus lower-index cladding supports total internal reflection at suitable angles (1). Any valid advantage, e.g. immunity to EM interference or high capacity (1).
6 [4]	Correct seven-layer order (1). Transport = process-to-process, ports (1); network = host-to-host/routing, IP (1); data link = next-hop frames, MAC (1).
7 [4]	FDM = frequency bands (0.5); WDM = optical wavelengths (0.5); TDM = time slots (1). Correct labelled sketch (1). Statistical slots have no permanently assigned channel position, so need identification (1).
8 [4]	UTP: twisted insulated copper without added shield (1). STP: shielding reduces interference when installed correctly (1). Fiber: light in core/cladding, immune to EM interference (1). Select fiber and justify with capacity/noise/distance (1).
9 [2]	Links = 8 × 7 / 2 = 28 (1). Ports per node = 8 - 1 = 7 (1).
10 [2]	There are 4 - 1 = 3 internal guards; B = 4 × 50 + 3 × 5 (1). B = 215 kHz (1).
11 [4]	Output = 4 × 2000 = 8000 bps = 8 kbps (1). Input bit = 1/2000 s = 0.5 ms (1). Output slot = 1/8000 s = 0.125 ms (1). Frame = 4 × 0.125 = 0.5 ms (1).

Repair the mistakes, then test again

For every lost mark: write the correct idea once, explain why your answer was wrong, then solve or answer it again without looking. Change the numbers in Q9-Q11 and repeat. Re-reading alone is not enough to check recall.

Extra numerical check

6 FDM channels × 20 kHz with 2-kHz internal guards → **130 kHz**.
5 inputs × 4 kbps, 1 bit/input/frame → **20 kbps**, **0.05 ms/slot**, **0.25 ms/frame**.
13 statistical output channels → **4 address bits**.

Sources and important clarifications

Source locations below use PDF slide/page numbers. All nine source files were supplied in D:\university\5th sem\Computer Networks (Theory)\notes. Administrative slides, repeated recaps and reading lists were condensed. Document assignments were treated as source content, not instructions to execute.

Source PDF	Coverage in this guide
CN-Lec-01 (Intro to data com).pdf	Pages 2-3; definitions, components, effectiveness, representation and digital chain.
CN-Lec-02 (Networks).pdf	Page 4; terms, modes, criteria and line configuration.
CN-Lec-03 (topologies).pdf	Pages 5-6; all six layouts, trade-offs and mesh calculations.
CN-Lec-04 (Categories of network).pdf	Page 7; scale, purpose, WAN types and internetworks.
CN-Lec-05 (OSI Model).pdf	Pages 8-9; layering, functions, interfaces and delivery.
CN-Lec-06 (TCP-IP Model).pdf	Page 10; model mapping, TCP/UDP, ports and supporting protocols.
CN-Lec-07 (guided media) (1).pdf	Pages 11-12; copper, fiber, categories, connectors and optical concepts.
CN-Lec-08 (unguided media).pdf	Pages 13-14; propagation, bands, antennas and medium choice.
CN-Lec-09 & 10 (Multiplexing).pdf	Pages 15-17; FDM, WDM, TDM, timing and rate management.

Read these before memorizing slide tables

Unicode (L1 p27): it is not always 32 bits; UTF-8/16/32 are different encoding forms. ASCII has 128 code positions (0-127). Unicode Consortium: encoding FAQ.

Cable rates (L7 p21): the table mixes historical figures with claims that should not be used as universal limits. Cat 5e supports Gigabit Ethernet. Cisco: cabling specifications.

RG-11 (L7 p28): the slide lists 50 ohms; standard RG-11 is 75 ohms. Do not rely on that row as a cable-selection rule. Belden: RG-11 cable.

Infrared (L8 p23): “400 GHz” is an upper-bound typo; the intended approximation is 400 THz, consistent with the following slide. NASA places infrared roughly between 1 mm and 700 nm (about 300 GHz to 430 THz). NASA: spectrum chart.

Other simplifications are clarified beside the relevant topic: bus failures, packet sizes, TCP versus UDP reliability, session-layer mapping, antenna patterns and fiber duplex operation. Historical examples are retained as lecture examples, not claims about current maximum performance. If your instructor tests a disputed slide value, ask which convention they expect.